

Generative Adversarial Networks (GANs)

- Goal
 - Train a generator to produce data indistinguishable from real data, while training a discriminator to distinguish real vs fake samples.
 - Essentially: G tries to fool D , D tries not to be fooled.
- Math:
 - Discriminator loss:
$$\mathcal{L}_D = -E_{x \sim p_{\text{data}}} [\log D(x)] - E_{z \sim p_z} [\log (1 - D(G(z)))]$$
 - Generator loss:
$$\mathcal{L}_G = -E_{z \sim p_z} [\log D(G(z))]$$
 - Optimization is minimax:
$$\min_G \max_D V(D, G) = E_{x \sim p_{\text{data}}} [\log D(x)] + E_{z \sim p_z} [\log (1 - D(G(z)))]$$
 - Training uses gradient descent/ascent on these losses.
- Conclusion:
 - GANs learn a probabilistic generator of data.
 - Training is unstable (loss oscillates, mode collapse can occur).
 - With enough iterations, G produces realistic samples similar to the real data distribution.